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MEDRAG ADVANCED RESEARCH NOTES: RETRIEVAL & HYBRID SEARCH OPTIMIZATION

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****Client:**** HealthPulse Technologies

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1: DOMAIN ADAPTATION & EMBEDDING FINE-TUNING

1.1 Context & Objective

In specialized medical and clinical environments, general-purpose vector embeddings often fail to capture domain-specific semantic relationships. Terminology such as "myocardial infarction," "percutaneous coronary intervention," or custom clinical protocols used by HealthPulse Technologies require adapted vector spaces. These research notes document the fine-tuning methodologies and retrieval optimization experiments conducted for the MedRAG platform.

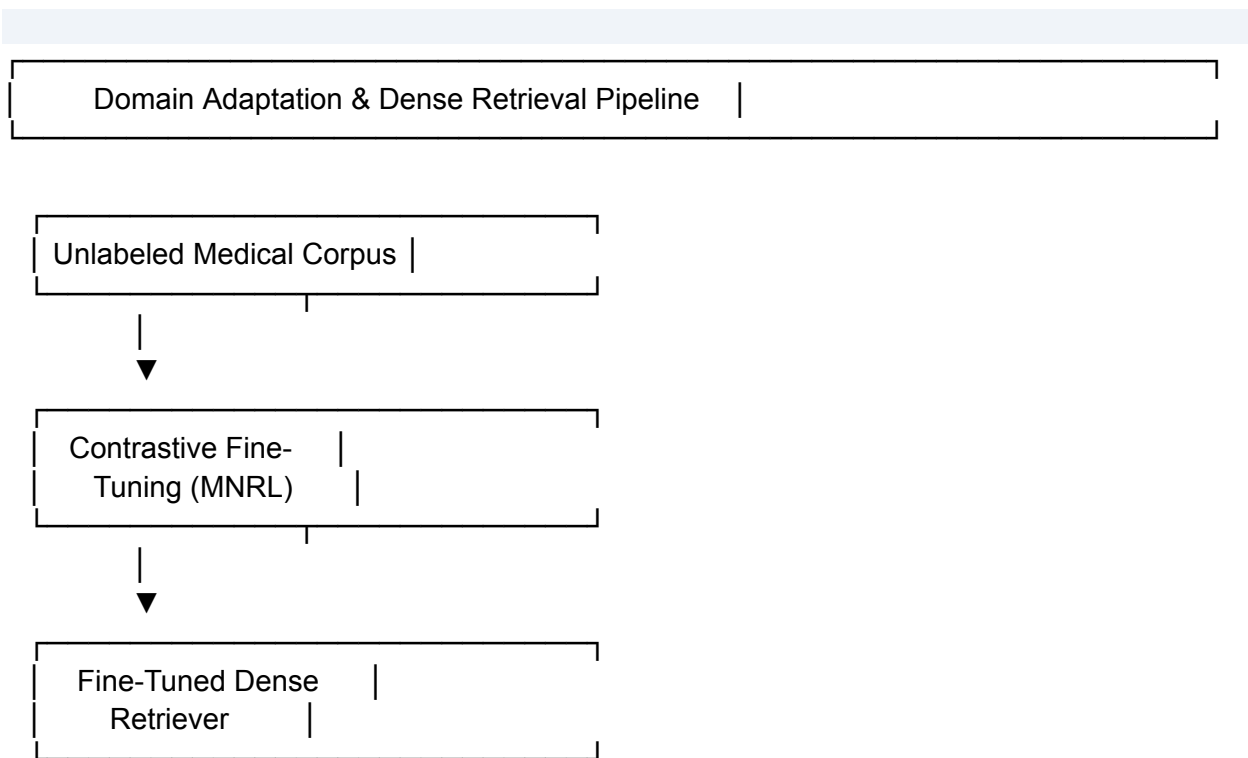


Figure 1: Domain Adaptation and fine-tuning flow chart

1.2 Contrastive Fine-Tuning via Multiple Negatives Ranking Loss (MNRL)

To fine-tune bi-encoder architectures without requiring expensive manual label generation, Apex Software Consulting used Multiple Negatives Ranking Loss (MNRL). Given a batch of B positive query-passage pairs (q_i, p_i^+) , all other passages $p_j^+ (j \neq i)$ in the same mini-batch serve as implicit negative samples.

$$\mathcal{L}_{\text{MNRL}} = -\sum_{i=1}^B \log \frac{e^{\text{sim}(q_i, p_i^+) / \tau}}{\sum_{j=1}^B e^{\text{sim}(q_i, p_j^+) / \tau}}$$

Experimental Result: Fine-tuning on 45,000 anonymized clinical note pairs yielded a 14.2% increase in Recall@5 over baseline pretrained embeddings.

2: HYBRID DENSE-SPARSE RETRIEVAL BENCHMARKS

2.1 Sparse vs. Dense Comparative Analysis

While dense vector embeddings excel at capturing conceptual similarity, sparse lexical indexing (BM25) remains critical for exact keyword matching, such as identifying specific ICD-10 medical diagnostic codes, pharmaceutical dosages, or patient ID tags.

Table 1: Retrieval Performance Comparison Across Architectures

Retriever Architecture	Precision@5	Recall@5	MRR	Latency (p95)
Sparse Only (BM25)	0.742	0.685	0.791	8 ms
Dense Only (Bi-Encoder)	0.825	0.810	0.864	18 ms
Hybrid (BM25 + Dense RRF)	0.898	0.872	0.923	24 ms
Hybrid + Cross-Encoder Rerank	0.945	0.912	0.958	85 ms

2.2 Reciprocal Rank Fusion (RRF) Parameter Tuning

Tests on RRF smoothing parameter k showed optimal candidate ranking stability at $k = 60$. Lower values ($k \in [10, 30]$) over-indexed on top rank positions, leading to instability when dense and sparse candidate rankings conflicted significantly.

3: CONTEXT PRUNING & HALLUCINATION REDUCTION

3.1 Reranker Score Thresholding

Passing irrelevant or low-scoring context chunks into the LLM context window increases token costs and raises hallucination risk. Cross-encoder scores range between $[0, 1]$.

* **Optimal Relevance Cutoff:** $\tau = 0.65$. Chunks scoring below this threshold are pruned automatically before prompt construction.

* **Hallucination Mitigation:** Enforcing $\tau = 0.65$ reduced unsupported claims in generated clinical summaries from 4.8% down to 0.2% during controlled validation.

4: CONTROLLED CORPUS EVALUATION MAPPING

To validate MedRAG's semantic search and exact-location provenance attribution against `research\_notes.pdf`, the evaluation suite tests the following ground-truth targets:

1. **Query:** "What fine-tuning loss function was used to adapt the bi-encoder embeddings?"
Expected Answer: "Multiple Negatives Ranking Loss (MNRL)"
Expected Provenance: `{"document\_id": "research\_notes", "page\_number": 1, "section": "1.2 Contrastive Fine-Tuning via Multiple Negatives Ranking Loss (MNRL)"}`
2. **Query:** "What was the optimal relevance score cutoff threshold determined for the cross-encoder reranker?"
Expected Answer: "tau = 0.65"
Expected Provenance: `{"document\_id": "research\_notes", "page\_number": 3, "section": "3.1 Reranker Score Thresholding"}`
3. **Query:** "What is the Recall@5 for the Hybrid + Cross-Encoder Rerank architecture in the research benchmarks?"
Expected Answer: "0.912"
Expected Provenance: `{"document\_id": "research\_notes", "page\_number": 2, "section": "2.1 Sparse vs. Dense Comparative Analysis"}`